

PHASE 1 — FOUNDATION

Week 7 — Capsule Build & First Splash

Build the capsules. Brief the deck. Every kid in the water.

Say Watt Robotics · ROV Team | 2026-2027 MATE ROV Season

MEETING 7 · THREE HOURS · POOL DAY

AGENDA

Three hours. Build the capsules. Brief the deck. Get wet.

0:00	Welcome & Safety Brief (Workshop Half)	10 min
0:10	Capsule Build — PVC, Caps, O-Rings, Epoxy	60 min
1:10	Travel to the Pool	20 min
1:30	Poolside Safety Brief	15 min
1:45	Every Kid in the Water — Comfort First	30 min
2:15	Smoke Test of the Control Box	15 min
2:30	Capsule Drop #1 + Towel Check	20 min
2:50	Wrap-Up at the Deck	10 min

Two safety briefs today — one before the workshop, one before the pool. No exceptions on either.

TODAY'S ROLES

Four jobs. Four students. Different every week.

Every meeting we assign these four — by season's end, every student has done every job.

NOTETAKER / SCRIBE

TODAY: _____

Captures every decision, action item, and blocker. Files notes with the CFO for the engineering notebook.

SAFETY BRIEFER

TODAY: _____

Delivers the standing four-question safety brief at the start of the meeting.

PHOTOGRAPHER

TODAY: _____

Captures photos and short videos throughout — feeds the engineering notebook and the marketing display.

TIMEKEEPER

TODAY: _____

Holds the meeting to the agenda's clock. Calls the time at each block change.

SECTION 01

Welcome & Safety Brief

0:00 - 0:10 · 10 MIN

Same ritual every meeting. We get sharper at it; the kids start running it.



Every meeting opens with a safety brief.

Why we do this

Real engineering and dive teams brief safety before every job — not because they expect to fail, but because the discipline of asking the same four questions every time is what makes a team safe in the long run. We start the habit now, when the stakes are small, so it's automatic when they aren't.

- 1 **What are we doing today?**
- 2 **What could hurt someone?**
- 3 **How do we prevent it?**
- 4 **Who do we tell if something happens?**

TODAY'S BRIEF · MEETING 7

THE WORK

Build watertight capsules: cut PVC, glue end caps with marine epoxy, set O-rings. Workshop half, then pool half.

THE HAZARDS

PVC primer and marine epoxy fumes. Sharp PVC edges after cutting. Pool half: drowning risk, pool-deck slips, sun.

PREVENTION

Workshop: ventilation on, gloves for epoxy, eye protection. Pool: lifeguard on duty, buddy system, dry hands at the deck edge.

IF SOMETHING HAPPENS

Workshop: stop, advisor first, then first-aid kit. Pool: kid in distress → lifeguard FIRST, advisor second. Buddy stays with kid.

LOOKING AHEAD

1.8 starts the Triggerfish frame and thruster integration. Capsule drop results inform whether we re-seal or move on.

SECTION 02

Build the Capsules

0:10 - 1:10 · 60 MIN

PVC, end caps, O-rings, marine epoxy. The capsule is the team's first watertight enclosure.



WHAT THE CAPSULE IS

The team's first watertight enclosure.

A capsule is a short length of PVC pipe with end caps sealed on. We drop it in the pool, retrieve it, and check whether the towel inside came out dry. The towel is the test — if it has even a damp spot, water got in, and the seal has to be redone.

Capsules are practice for the real watertight enclosure on the TriggerFish — the WTC that holds the control board. Same physics, smaller stakes. If we can't keep water out of a sealed PVC tube, we can't keep it out of a \$300 enclosure with a \$400 Raspberry Pi inside.

Phase 1's exit gate (1.11) is three consecutive dry dives. Today is dive #1. The work tonight starts the count.

Six steps. Marine epoxy is the long-pole item.

1

Cut the PVC to length

Pipe cutter or hacksaw, then deburr both ends with sandpaper. Sharp inner edges cut O-rings.

2

Dry-fit the end caps

Both ends. Caps should slide on snug but not require a hammer. If they're loose, the capsule will leak; if too tight, the O-rings won't seat.

3

Set the O-rings

Lubricate the O-ring lightly with silicone grease. Seat it in the cap's groove. Inspect: no twists, no pinches, fully seated all the way around.

4

Mix the marine epoxy

Equal parts A and B, mix for 45 seconds. Working time after mix is about 5 minutes. Don't mix until both caps are ready to set.

5

Glue one cap, set, wait, glue the other

Apply epoxy to one cap, press onto the pipe, hold 30 seconds. Set the capsule down, end-cap up. Wait 15 minutes before the second cap so the first sets in tension, not compression.

6

Insert a dry towel before the second cap goes on

The towel is the test. Quartered washcloth fits a 4-inch capsule cleanly. After the towel is in, glue and set the second cap.

SECTION 03

Poolside Safety Brief

1:30 - 1:45 · 15 MIN

Pool deck rules. Lifeguard introduction. Buddy assignment. No water touches us yet.



Eight rules. Every pool session. Every time.

Different from the workshop safety brief. We re-do this every time we're at a pool — even when the kids think they remember.

1

Lifeguard is the safety authority on deck

If the lifeguard says move, we move. If they say out of the water, we get out. No discussion.

2

Buddy system, always

Every kid has a named buddy. You stay within arm's reach of your buddy when in the water. If your buddy gets out, you get out.

3

Dry hands at the deck edge

Wet hands on a wet deck on a wet kid is how someone goes down. Towel off before standing at the edge.

4

Tether stays on the deck, never around a kid

The team's tether and any other line stays coiled on the deck, never wrapped or draped on a swimmer.

5

Walk, don't run, on the deck

Always. The pool deck is the slipperiest surface most kids encounter in a week.

6

Hydration

Water bottles on the deck, accessible. Pool air dehydrates faster than people expect.

SECTION 04

Every Kid in the Water

1:45 - 2:15 · 30 MIN

Phase 1 exit gate. Comfort and familiarity first — no missions, no tasks, just being in the pool with the team.



THE WHY

Comfort first. Tasks come later.

Tonight is not a mission. It's not a test. It's every kid getting wet with the team so the water stops being a foreign environment by April. Phase 1's exit gate includes this beat explicitly: every member has been in the water before the season transitions to Phase 2.

Buddy assignments are made on the deck. Stronger swimmers buddy weaker swimmers; nobody is alone. The advisor and the lifeguard each track which buddies are in and which are out.

No ROV in the water tonight. No tether in the water tonight. The team gets in, swims around, gets out, dries off. We learn the deck.

SECTION 05

Smoke Test of the Control Box

2:15 - 2:30 · 15 MIN

We are NOT in the water for this. Smoke test happens on the deck, with the battery, with one person hand on the disconnect.



Battery to board. Three readings. Disconnect.

If the meter-check from 1.6 holds, this is uneventful. That's the goal.

- 1 Re-verify the meter-check from 1.6**
Quick pass with the multimeter on every pad — same three reads. If anything has shifted since 1.6, fix before connecting power.
- 2 Connect the battery — slowly**
Polarity verified by two people. Battery connected at the watt meter end first, then the board end. One person watches the watt meter; one person has a hand on the disconnect.
- 3 Read the watt meter for ~5 seconds**
Expected: small idle current, no smoke, no smell, no audible pop. Anything else → disconnect immediately.
- 4 Disconnect, document the result**
Idle current reading + any anomalies → engineering notebook. Signed and timed. If clean, we move to capsule drop #1.

SECTION 06

Capsule Drop #1 & Towel Check

2:30 - 2:50 • 20 MIN

*First of three consecutive dry dives needed for the Phase 1 exit gate.
Today is the start of the count.*



Five moves. The towel decides.

1

Drop the capsule into the deepest available point

Underhand toss from the deck. Capsule sinks if weighted correctly; floats if not — note which and adjust before drop #2.

2

Retrieve after 60 seconds

Pool noodle or a kid with a buddy retrieves. Capsule comes out wet on the outside — that's expected.

3

Dry the outside before opening

Wipe down the capsule completely before cracking either end cap. We don't want to introduce water during the inspection.

4

Open the leading end cap. Inspect the towel.

Bone-dry towel = pass. Damp towel = fail. Even a spot. Capsule fails, the seal gets redone before drop #2.

5

Log the result in the engineering notebook

Date, time, capsule ID (if more than one), pass/fail, photo of the towel. This log is the proof we hit the three-consecutive-dry-dives gate at 1.11.

SECTION 07

Wrap-Up & Homework

2:50 - 3:00 · 10 MIN

Capture what we decided. Close the loop. Hand off the next move.



Before we meet again.

EVERY STUDENT

- 1 If capsule drop #1 was a fail, the team that built it owns the re-seal — done before 1.8 starts.
- 2 Photographer: upload pool-deck and capsule photos to the team folder TONIGHT — sun-dried colors fade fast in the camera roll.
- 3 Bring closed-toe shoes for 1.8 — back to the workshop for the Triggerfish frame and thruster integration.

THE PARALLEL TRACK · LEADS & ADVISORS

Advisors: thank the lifeguard and the host individually before leaving. Confirm pool access for the rest of Phase 1 — 1.9 (capsule dive #2) needs the same host. If host availability is shaky, lock the deep-water alternate now.

NEXT MEETING · WEEK 8

Build & Iterate I

Triggerfish frame, thrusters, tether — the first full build. Pool ops and deck discipline carry forward.

ACTION ITEMS

What we decided. Who owns it. When it's due.

The Notetaker fills these in during the meeting and reads them back before we leave.

ITEM	OWNER	DUE

Tools accounted for. Shop reset.

We do this together, every meeting. No exceptions — the Tool Steward leads.

- ☐ All capsules accounted for; failed ones bagged for re-seal at home.
- ☐ Engineering notebook entries signed for smoke test and capsule drop.
- ☐ Battery DISCONNECTED, secured, transported back to the workshop.
- ☐ Multimeter, watt meter, towels, first-aid kit packed.
- ☐ Pool deck swept; team area returned cleaner than we found it.
- ☐ Photos uploaded to the team folder tonight, not tomorrow.
- ☐ Thank-you to the lifeguard and the pool host delivered in person.

EVERY KID WET. FIRST DIVE LOGGED.

The pool is not foreign anymore.

Two safety briefs delivered. Smoke test clean. Drop #1 logged. The Phase 1 gate is in sight.